



East West University
Department of Computer Science & Engineering
Course Outline of CSE360/ICE469
Spring 2025 Semester

Course Information

Course: CSE360 Computer Architecture

Teaching Scheme:

	Theory	Laboratory	Total
Credit Hours	3	0	3
Contact Hours	2.5 Hours/Week for 15 Weeks + Final Exam in the 16th week	0	2.5 Hours/Week for 15 Weeks + Final Exam in the 16th week

Prerequisite: CSE325 Operating Systems, CSE345 Digital Logic Design

Instructor Information

Instructor: Dr. Md. Nawab Yousuf Ali
Professor, Department of Computer Science and Engineering
Office: Room # 644
E-mail: nawab@ewubd.edu
URL: <http://www.ewubd.edu/~nawab/-personal/>
TA: TBA

Class Routine and Office Hour

Days	Class Hours	Office Hour	Course & sections	Room no.
Sunday		11:50-01:20		
Monday	10:10-11:40	11:50-01:20 01:30-03:00	CSE360(1) ICE469(1)	226
Tuesday		11:50-01:20		
Wednesday	10:10-11:40	11:50-01:20 01:30-03:00	CSE360(1) ICE469(1)	226
Thursday				

Course Objective

The objective of this course is to study the structure, behaviour, and characteristics of computer systems. This course will exhibit the design of the various functional units of digital computers, discuss different types of memories and their properties, and introduce basics of parallel computer architecture. Knowledge of this course will be needed as prerequisite knowledge for future course, such as CSE442 Microprocessors and Microcontrollers.

Knowledge Profile

K3: Theory-based engineering fundamentals

Learning Domains

Cognitive – C2: Understanding, C3: Applying, C4: Analyzing

Program Outcomes (POs)

PO1: Engineering Knowledge

PO2: Problem Analysis

Complex Engineering Problem Solution

EP1: Depth of knowledge required.

EP2: Range of conflicting requirements

EP3: Depth of analysis required.

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	Complex Engineering Problem Solving/ Engineering Activities
CO1	Understand the structure, function, and characteristics of different functional units of digital computers and apply methods of implementing them to have an overall perspective of digital computers.	PO1	C2, C3	K3	
CO2	Understand , and determine performance of memory and I/O subsystems to synthesize functional unit of digital computers.	PO1	C2, C3	K3	

CO3	Understand, implement, and examine instruction set design for performance improvement, demonstrate this knowledge, and write report for effective design of digital computers.	PO2	C2, C3, C4 P1, P2	K3	EP1, EP2, EP3
CO4	Understand, Implement, and examine processor and control unit design, demonstrate this knowledge, and write report to synthesize functional units of digital computers.	PO2	C2, C3, C4 P1, P2	K3	EP1, EP2, EP3

Course Topics, Teaching-Learning Method, and Assessment Scheme

Course Topic	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			Mark of Psychomotor Learning Levels		CO Mark	Exam (Mark)
			C2	C3	C4	P1	P2		
Basic Concepts: Function and structure of a computer, Functional components of a computer, Interconnection of components, Performance analysis of a computer, Hardware architecture	Lecture, Class Discussion, Discussion Outside Class with Instructor/Teaching Assistant	CO1	3	3				6	Midterm Assessment I (30)
Representation of Instructions: Bus Interconnection, Scalar Data Types, Fixed and Floating-point numbers, Signed numbers, Integer Arithmetic, 2's Complement method for multiplication, Booths Algorithm,		CO1		10	4			14	

Floating point representations, IEEE standards, Floating point arithmetic									
Memory Organization Characteristics of memory systems, Internal and External Memory, Types of memories: ROM: PROM, EPROM, EEPROM, RAM: SRAM, DRAM, SDRAM, RDRAM, High-Speed Memories: Cache Memory, elements of cache design, Pentium 4 cache, Organization and Mapping Techniques, Replacement Algorithms, Cache Coherence		CO2	5	5				10	
Secondary Storage: Magnetic Disk (Hard Disk), Tape, DAT, RAID, Optical memory, CDROM, DVD, Error correction memories, Interleaved memories, Hardware support of memory management		CO2		5				5	Final Exam (40)
Input/Output Organization: External Devices, I/O modules, Programmed I/O, Interrupt-driven I/O,		CO2		5				5	

Direct memory access, I/O channels and processors, external interface									
Instruction Sets: Machine instruction, operands, operations, and assembly language, addressing, and instruction format		CO3	5	5	5			15	
CPU structure and function: processor and registers organization, instruction cycle and instruction pipelining, Reduced Instruction Set Computers (RISC), superscalar processors, parallel processing, Micro programmed control unit		CO4	5	5	5			15	

Mini Project

Course Topic	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			Mark of Psychomotor Learning Levels		CO Mark	Exam (Mark)
			C2	C3	C4	P1	P2		
Mini Project	Team-based moderately complex project with report writing and oral presentation	CO3	1	1	1	1	1	5	Mini project (10)
		CO4	1	1	1	1	1	5	

Overall Assessment Scheme

Assessment Area	CO				Assessment Area Mark
	CO1	CO2	CO3	CO4	
Class Test/Quiz	2.5	2.5	2.5	2.5	10
Midterm Assessment	20.0	10.0			30
Final Exam		10.0	15.0	15.0	40
Mini Project			5.0	5.0	10
Assignment/Presentation			5.0	5.0	10
Total Mark	22.5	22.5	27.5	27.5	100

Teaching Materials/Equipment

Textbook:

[1] *William Stallings*, Computer Organization and Architecture: Designing for Performance, Eleventh Edition, Pearson, 2023.

Mini Project:

Mini Project will be provided.

Grading System

Marks (%)	Letter Grade	Grade Point	Marks (%)	Letter Grade	Grade Point
80-100	A+	4.00	55-59	B-	2.75
75-89	A	3.75	50-54	C+	2.50
70-74	A-	3.50	45-49	C	2.25
65-69	B+	3.25	40-44	D	2.00
60-64	B	3.00	0-39	F	0.00

Exam Dates

Section	Mid Term Assessment	Final
1	Will be announced	26 May 2025

Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offence under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student **MUST** have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter into the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss any exam, the student **MUST** get approval of makeup exam by written application to the Chairperson through the Course Instructor **within 48hours** of the exam time. Proper supporting documents in favor of the reason of missing the exam have to be presented with the application.
- For **the final exam**, there will be NO makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses the final exam, the student **MUST** get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.**
- All mobile phones **MUST** be turned to silent mode during class and exam periods.
- There is **zero tolerance for cheating** in exams. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university.**